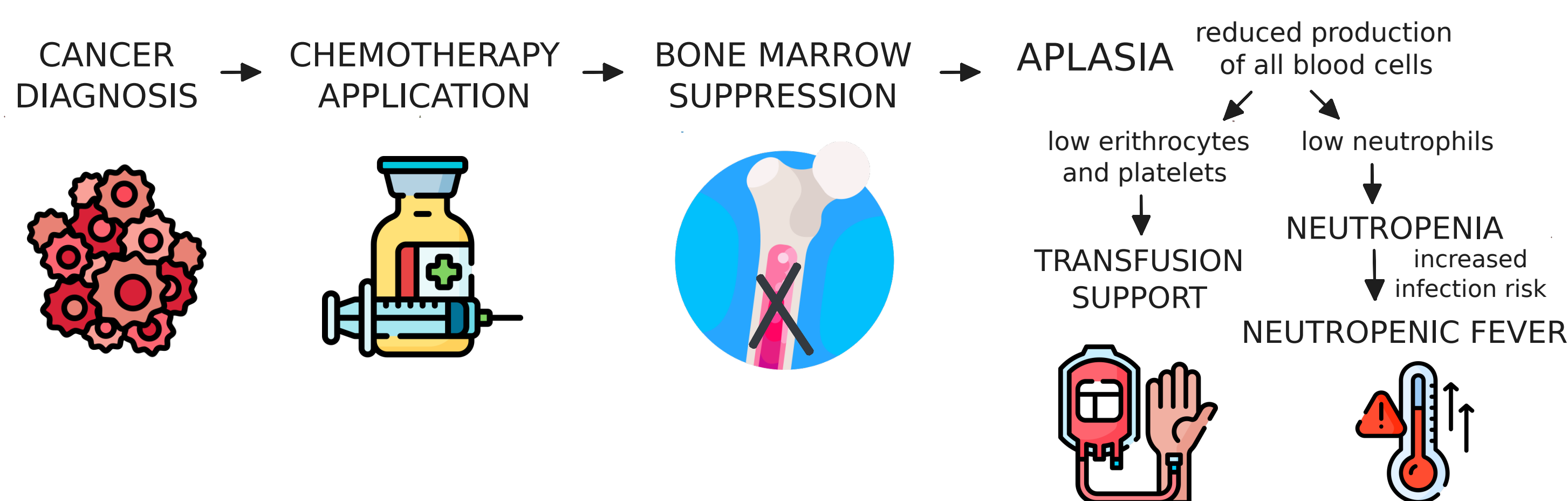




Research Objective

Background: Chemotherapy is a widely used cancer treatment that targets rapidly dividing cells. However, it can also harm healthy tissue, leading to **bone marrow suppression** and hematologic complications [1], including:

- **Aplasia** - failure of the bone marrow to produce sufficient blood cells, resulting in critical deficiencies in one or more cell lines;
- **Neutropenic Fever (NF)** - fever in patients with markedly reduced neutrophil count associated with a high risk of infection.



Objective: Develop predictive models to identify patients at risk of neutropenic fever or aplasia before these complications arise.

Datasets and Cohort Definition

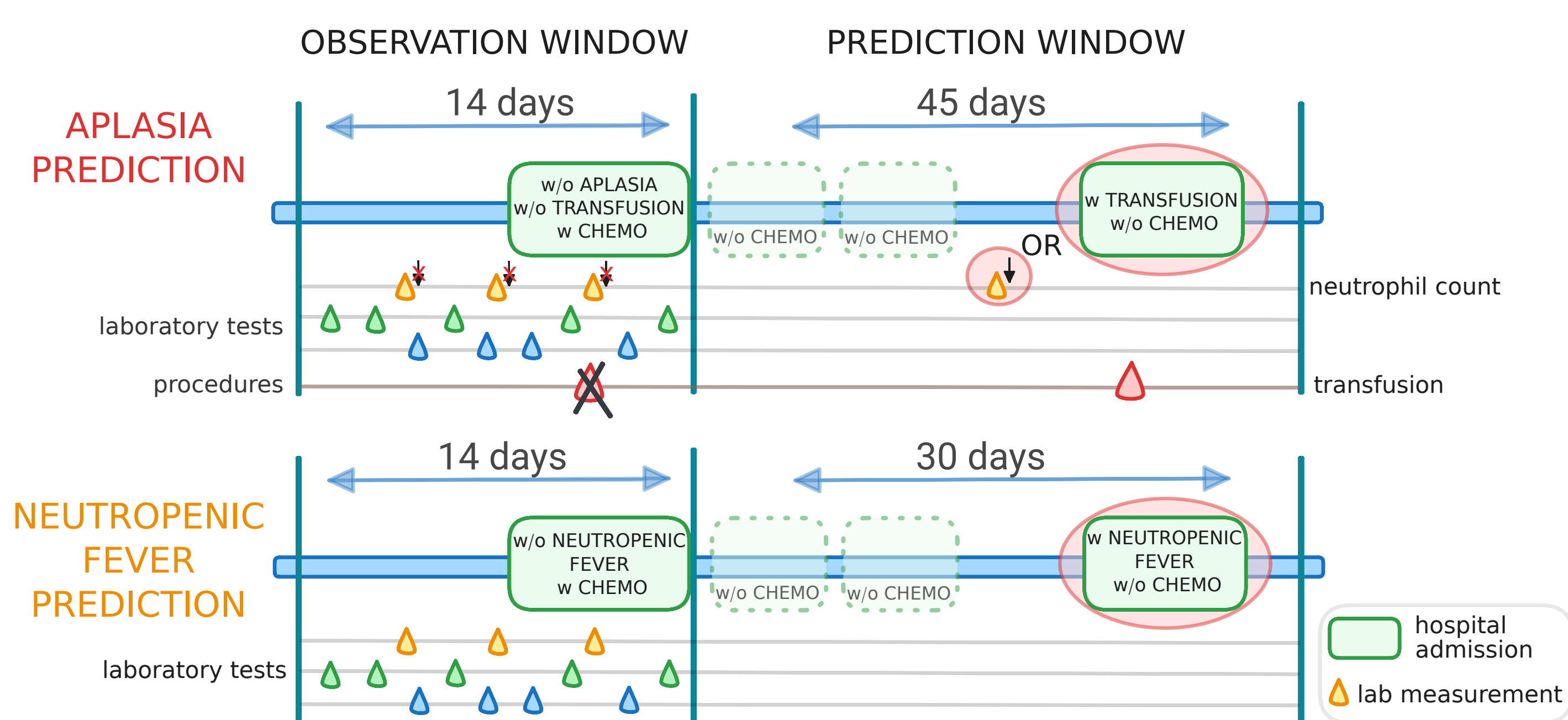
Datasets. Electronic health records from the public MIMIC-IV database [2] and a pediatric oncology dataset from Uniklinikum Erlangen (UKER).

Selected patients. Individuals who received any cancer diagnosis and at least one chemotherapy treatment.

Cohorts.

- **Aplasia:** admissions without aplasia, with aplasia identified by transfusion OR absolute neutrophil count $< 500/\mu\text{L}$
Target » aplasia onset within **45 days** post-discharge
- **Neutropenic fever:** admissions without NF diagnosis, identified by ICD codes.
Target » NF diagnosis within **30 days** post-discharge.

Observation window. 14 days preceding discharge.



Feature Extraction

Feature Types:



Laboratory measurements (14 days prior to discharge)



Patient demographics (age and gender)

- **Feature selection:** In UKER, all 48 recorded temporal features were used. In MIMIC-IV, the 100 most frequently recorded temporal features among positive cases were selected to mitigate data sparsity.

Machine Learning Framework

- **Class imbalance:** strong imbalance handled via minority oversampling and class-weighted loss functions.
- **Missing values:** handled by forward-fill and backward-fill, followed by mean imputation for any remaining missing values.
- **Models:** divided into non-sequential and sequential based on whether they account for temporal patterns and can handle irregular time series.
 - **Non-sequential:** Logistic Regression (LR), and **tree-based models** Random Forest (RF), Gradient Boosting (GB), and eXtreme Gradient Boosting (XGB).
 - **Sequential (regular series):** Gated Recurrent Unit (GRU), Long Short-Term Memory (LSTM), Temporal Convolutional Network (TCN), and Simply Attend and Diagnose (SAnD), trained on regularly sampled time series with added missingness and time-since-last-observed indicators [3].
 - **Sequential (irregular series):** Interpolation Network (IntNet), GRU-D, and Self-Supervised Transformer for Time Series (STraTS).
- **Training:** 5-fold cross-validation, stratified by patient.
- **Evaluation metrics:** Area under the ROC curve (ROC), area under the precision-recall curve (PR), and F1-score (F1).

Results

Model	APLASIA						NEUTROPENIC FEVER					
	MIMIC-IV			UKER			MIMIC-IV			UKER		
	ROC	PR	F1	ROC	PR	F1	ROC	PR	F1	ROC	PR	F1
Non-sequential models												
LR	0.838	0.574	0.599	0.753	0.601	0.580	0.727	0.128	0.205	0.643	0.105	0.168
RF	0.876	0.654	0.652	0.832	0.709	0.656	0.801	0.216	0.252	0.780	0.171	0.166
GB	0.881	0.688	0.637	0.847	0.743	<u>0.665</u>	<u>0.799</u>	0.228	0.167	0.771	0.178	0.094
XGB	<u>0.879</u>	<u>0.684</u>	<u>0.648</u>	<u>0.842</u>	<u>0.730</u>	0.672	0.792	0.212	0.209	0.757	<u>0.179</u>	0.122
Sequential models (regular time series)												
GRU	0.868	0.649	0.626	0.824	0.678	0.640	0.792	0.174	0.204	<u>0.783</u>	0.169	0.230
LSTM	0.864	0.634	0.625	0.803	0.655	0.615	0.766	0.160	0.196	0.766	0.163	0.228
SAnD	0.859	0.624	0.600	0.828	0.692	0.604	0.763	0.172	<u>0.220</u>	0.778	0.171	<u>0.245</u>
TCN	0.874	0.662	0.643	0.832	0.700	0.653	0.786	0.198	0.208	0.790	0.188	0.248
Sequential models (irregular time series)												
IntNet	0.741	0.486	0.518	0.719	0.547	0.533	0.672	0.096	0.161	0.591	0.076	0.129
GRU-D	0.869	0.658	0.635	0.834	0.702	0.662	0.782	0.184	0.195	0.778	0.168	0.233
STraTS	0.861	0.637	0.557	0.828	0.680	0.620	0.751	0.159	0.159	0.782	0.167	0.216
											Best score	
											Second-best score	

Conclusions

- Classical and temporal machine learning models can **effectively predict patients at risk** of aplasia and neutropenic fever prior to clinical onset using routinely collected laboratory data.
- **Tree-based models** achieved performance at least comparable to that of temporal deep learning models, suggesting that fine-grained temporal patterns may not offer additional predictive value in this context.

References

- [1] N. Behranvand et al., "Chemotherapy: A double-edged sword in cancer treatment," *Cancer Immunology, Immunotherapy*, vol. 71, no. 3, pp. 507–526, 2022.
- [2] A. E. Johnson et al., "Mimic-iv, a freely accessible electronic health record dataset," *Scientific Data*, vol. 10, no. 1, p. 1, 2023.
- [3] Z. Che et al., "Recurrent neural networks for multivariate time series with missing values," *Scientific reports*, vol. 8, no. 1, p. 6085, 2018.